



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO S 11043 O SOWDAPURAM PACCS
LIMITED**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

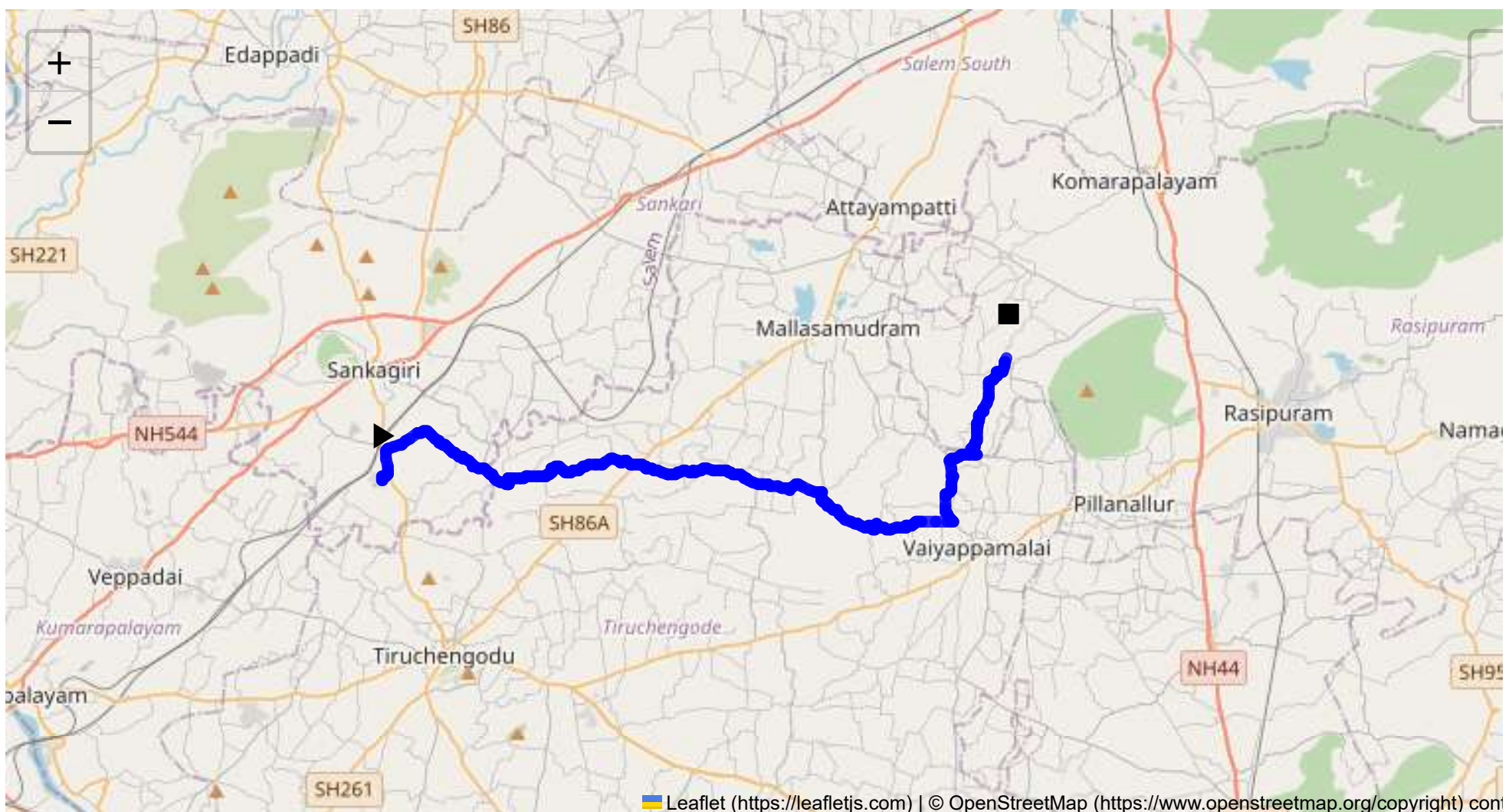
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 34.20 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.48009411, 78.09094964)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route stretches approximately 34.20 kilometers, passing through multiple regions in Tamil Nadu, India. Starting from Sangagiri, it moves through Kangeyampalayam, Morepalayam, Morangam, and Thottiyappatti, finally reaching Sowthapuram. The journey mainly involves traversing district and state highways, characterized by rural and semi-urban landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a hot and humid climate, with significant influence from the North-East and South-West monsoons. This region might face heavy rainfall from June to September and occasionally in the later monsoon season. Flash floods or waterlogging can occur in areas with poor drainage systems, affecting traffic flow and visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, traffic may peak during morning (7:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) hours, especially near urban areas and market zones.
- **Congestion-Prone Areas:** Key congestion points may include intersections near Tiruchengode and market areas around Pillanatham.

4. Assessment of Road Quality and Infrastructure

Road conditions can vary along this route. Main highways are generally well-maintained, but smaller roads leading to villages may have potholes or be narrower, requiring careful navigation.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, detour options include using the NH544 for stretches that run parallel or adjacent to the route path, provided the vehicle can handle the potential increase in distance and travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires compliance with the Motor Vehicles Act and specific state regulations, typically involving:

- Display of appropriate hazard placards
- Specialized licensing for the driver
- Restricted travel during peak city hours

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific historical data isn't available without accessing local records, national highways in India are known for occasional incidents involving heavy vehicles. Factors often implicated include speeding, mechanical failures, and driver fatigue.

8. Environmental Considerations and Sensitive Areas

The route may cross agricultural zones and small forest patches. It's vital to minimize any environmental impact, particularly concerning potential chemical spills from heavy vehicles.

9. Analysis of Communication Coverage

Mobile network coverage is relatively stable along this route, but some rural stretches may experience weak signals or brief dead zones, especially in hilly or densely vegetated areas.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times can vary:

- Near urban or semi-urban regions (e.g., Sangagiri, Tiruchengode): Approximately 15-30 minutes
- Remote or rural segments may take longer, ranging from 30-60 minutes due to limited infrastructure.

12. Overall Summary of Risk Assessment

The overall risk assessment of this route indicates a moderate level of risk. Risks arise from variable road quality, peak-hour congestion, potential poor weather conditions, and moderate emergency response times. Planning for alternate routes and effective communication are crucial for safe passage, especially when transporting hazardous materials. The route can be safely navigated by adhering to local regulations and maintaining vigilance in identified risk areas.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44028, 77.87541	15 KM/Hr
3	Turn	High	11.44861, 77.87429	15 KM/Hr
4	Turn	High	11.43673, 77.91697	15 KM/Hr
5	Turn	High	11.43679, 77.91705	15 KM/Hr
6	Turn	High	11.43922, 77.91702	15 KM/Hr
7	Turn	Medium	11.43928, 77.91712	30 KM/Hr
8	Turn	Medium	11.43939, 77.92306	30 KM/Hr
9	Turn	Medium	11.43965, 77.92336	30 KM/Hr
10	Turn	Medium	11.43996, 77.92595	30 KM/Hr
11	Turn	Medium	11.44025, 77.93057	30 KM/Hr
12	Turn	High	11.44080, 77.93220	15 KM/Hr
13	Turn	Medium	11.44135, 77.93220	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.44154, 77.93238	30 KM/Hr
15	Turn	Medium	11.44200, 77.93249	30 KM/Hr
16	Turn	Medium	11.44234, 77.93242	30 KM/Hr
17	Turn	High	11.44248, 77.93247	15 KM/Hr
18	Turn	Medium	11.44287, 77.93430	30 KM/Hr
19	Turn	Medium	11.44277, 77.94291	30 KM/Hr
20	Turn	Medium	11.44549, 77.95319	30 KM/Hr
21	Turn	Medium	11.44471, 77.95806	30 KM/Hr
22	Turn	Medium	11.44389, 77.96281	30 KM/Hr
23	Turn	High	11.44395, 77.96298	15 KM/Hr
24	Turn	Medium	11.44296, 77.96485	30 KM/Hr
25	Turn	High	11.44240, 77.96502	15 KM/Hr
26	Turn	Medium	11.44012, 77.97524	30 KM/Hr
27	Turn	Medium	11.44104, 77.98003	30 KM/Hr
28	Turn	Medium	11.44163, 77.98464	30 KM/Hr
29	Turn	Medium	11.44022, 77.99819	30 KM/Hr
30	Turn	Medium	11.43693, 78.00596	30 KM/Hr
31	Turn	Medium	11.43523, 78.01510	30 KM/Hr
32	Turn	Medium	11.43534, 78.01541	30 KM/Hr
33	Turn	Medium	11.43720, 78.01776	30 KM/Hr
34	Turn	Medium	11.43442, 78.02639	30 KM/Hr
35	Turn	High	11.43435, 78.02651	15 KM/Hr
36	Turn	Medium	11.43059, 78.02784	30 KM/Hr
37	Turn	Medium	11.42467, 78.03438	30 KM/Hr
38	Turn	High	11.42326, 78.03854	15 KM/Hr
39	Turn	High	11.42363, 78.03878	15 KM/Hr
40	Turn	Medium	11.42364, 78.03893	30 KM/Hr
41	Turn	Medium	11.42218, 78.04120	30 KM/Hr
42	Turn	High	11.42219, 78.04286	15 KM/Hr
43	Turn	High	11.42262, 78.04311	15 KM/Hr
44	Turn	Medium	11.42248, 78.04345	30 KM/Hr
45	Turn	High	11.42150, 78.04464	15 KM/Hr
46	Turn	Medium	11.42164, 78.04483	30 KM/Hr
47	Turn	High	11.42319, 78.04543	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
48	Turn	Medium	11.42117, 78.05023	30 KM/Hr
49	Turn	Medium	11.42220, 78.05293	30 KM/Hr
50	Turn	Medium	11.42190, 78.05555	30 KM/Hr
51	Turn	High	11.42203, 78.05577	15 KM/Hr
52	Turn	High	11.42259, 78.05580	15 KM/Hr
53	Turn	Medium	11.42274, 78.05597	30 KM/Hr
54	Turn	Medium	11.42444, 78.06039	30 KM/Hr
55	Blind Spot	Blind Spot	11.42430, 78.07218	10 KM/Hr
56	Turn	Medium	11.43346, 78.06964	30 KM/Hr
57	Turn	Medium	11.43409, 78.07098	30 KM/Hr
58	Turn	High	11.44484, 78.07074	15 KM/Hr
59	Turn	Medium	11.44694, 78.07489	30 KM/Hr
60	Blind Spot	Blind Spot	11.44688, 78.08030	10 KM/Hr
61	Turn	Medium	11.44745, 78.07950	30 KM/Hr
62	Blind Spot	Blind Spot	11.44930, 78.07877	10 KM/Hr
63	Turn	Medium	11.44998, 78.07988	30 KM/Hr
64	Turn	Medium	11.45595, 78.08019	30 KM/Hr
65	Turn	Medium	11.45627, 78.08032	30 KM/Hr
66	Turn	Medium	11.45650, 78.08058	30 KM/Hr
67	Turn	High	11.45681, 78.08071	15 KM/Hr
68	Turn	High	11.45732, 78.08038	15 KM/Hr
69	Turn	High	11.46189, 78.08268	15 KM/Hr
70	Turn	Medium	11.47157, 78.08461	30 KM/Hr
71	Turn	Medium	11.47179, 78.08475	30 KM/Hr
72	Turn	Medium	11.47426, 78.08763	30 KM/Hr
73	Blind Spot	Blind Spot	11.47540, 78.08765	10 KM/Hr
74	Turn	High	11.47566, 78.08963	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Sri Vinayaga Matriculation Higher Secondary School	11.4233682, 78.0736517	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44861, 77.87429



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43673, 77.91697



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43679, 77.91705



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43922, 77.91702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43928, 77.91712



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43939, 77.92306



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43965, 77.92336



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43996, 77.92595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44025, 77.93057



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44080, 77.93220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44135, 77.93220



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44154, 77.93238



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44200, 77.93249



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44234, 77.93242



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44248, 77.93247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44287, 77.93430



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44277, 77.94291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44549, 77.95319



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44471, 77.95806



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44389, 77.96281



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44395, 77.96298



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44296, 77.96485



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44240, 77.96502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44012, 77.97524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44104, 77.98003



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.98464



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44022, 77.99819



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43693, 78.00596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43523, 78.01510



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43534, 78.01541



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43720, 78.01776



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43442, 78.02639



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43435, 78.02651



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43059, 78.02784



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42467, 78.03438



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42326, 78.03854



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42363, 78.03878



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42364, 78.03893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42218, 78.04120



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42219, 78.04286



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42262, 78.04311



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42248, 78.04345



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42150, 78.04464



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42164, 78.04483



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42319, 78.04543



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42117, 78.05023



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42220, 78.05293



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42190, 78.05555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42203, 78.05577



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42259, 78.05580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42274, 78.05597



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42444, 78.06039



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42430, 78.07218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43346, 78.06964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43409, 78.07098



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44484, 78.07074

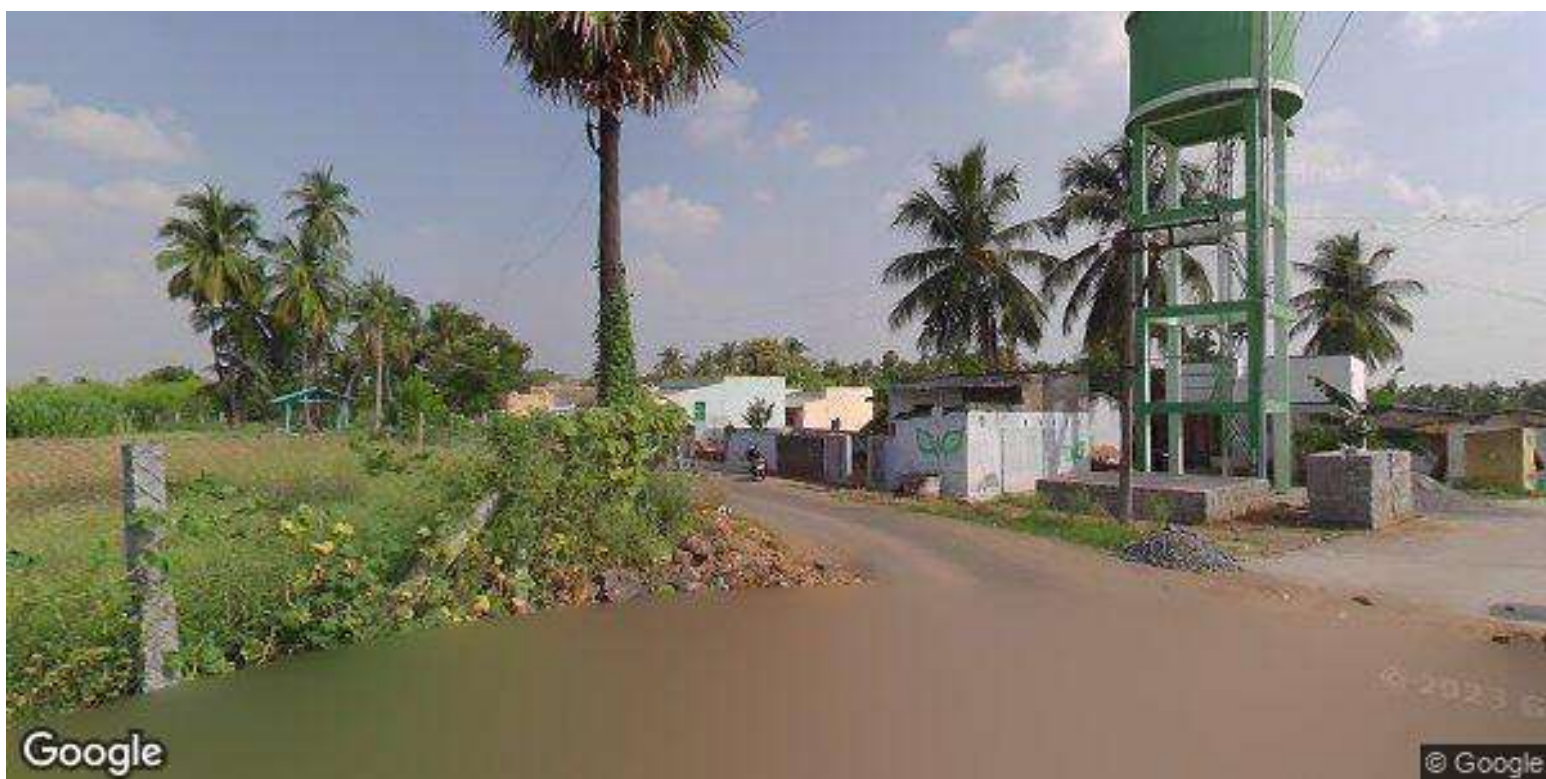


Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44688, 78.08030



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

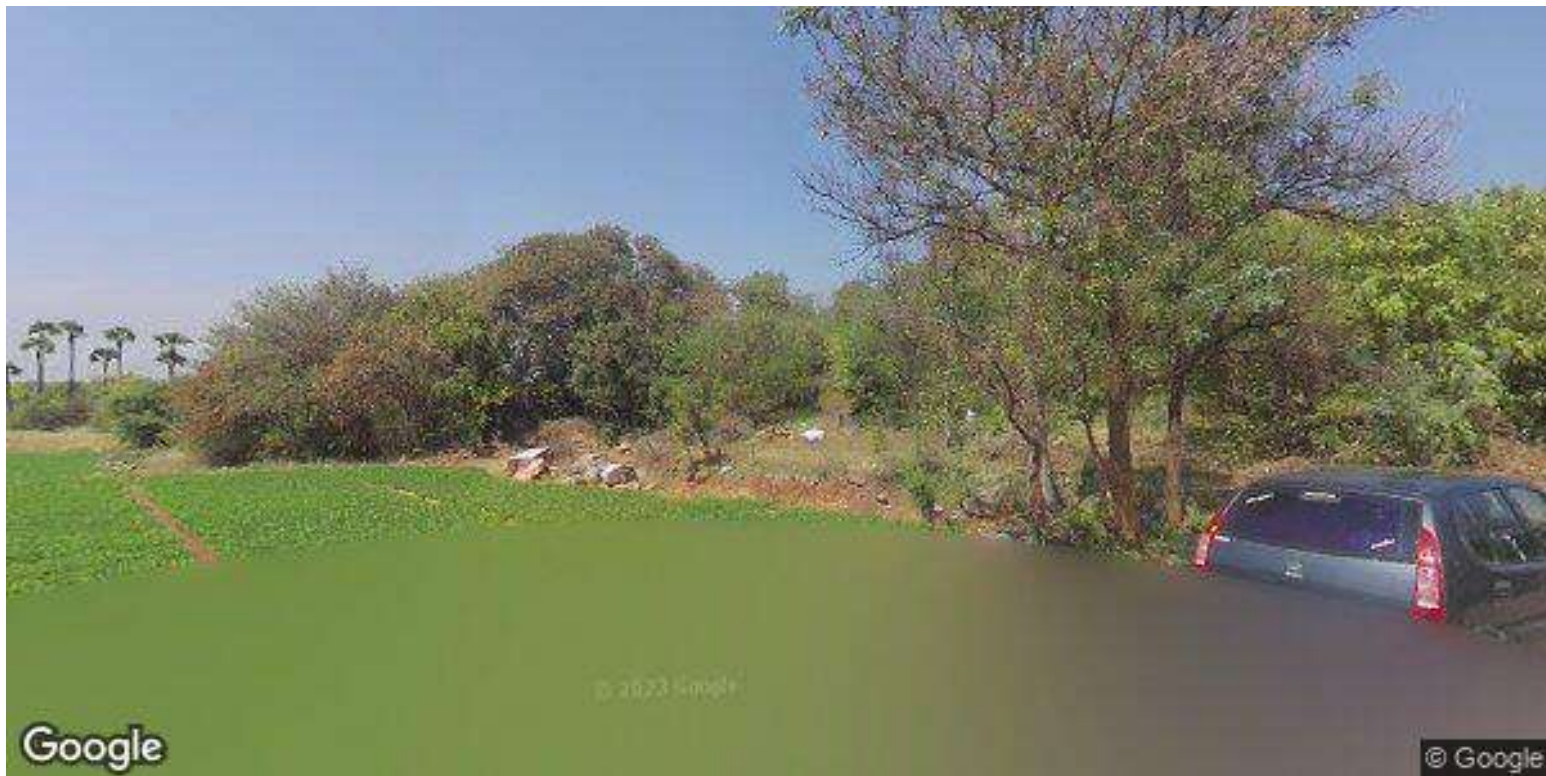
Coordinates: 11.44745, 78.07950



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44930, 78.07877



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44998, 78.07988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45595, 78.08019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45627, 78.08032



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45650, 78.08058



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45681, 78.08071



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45732, 78.08038



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46189, 78.08268



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47157, 78.08461



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47179, 78.08475

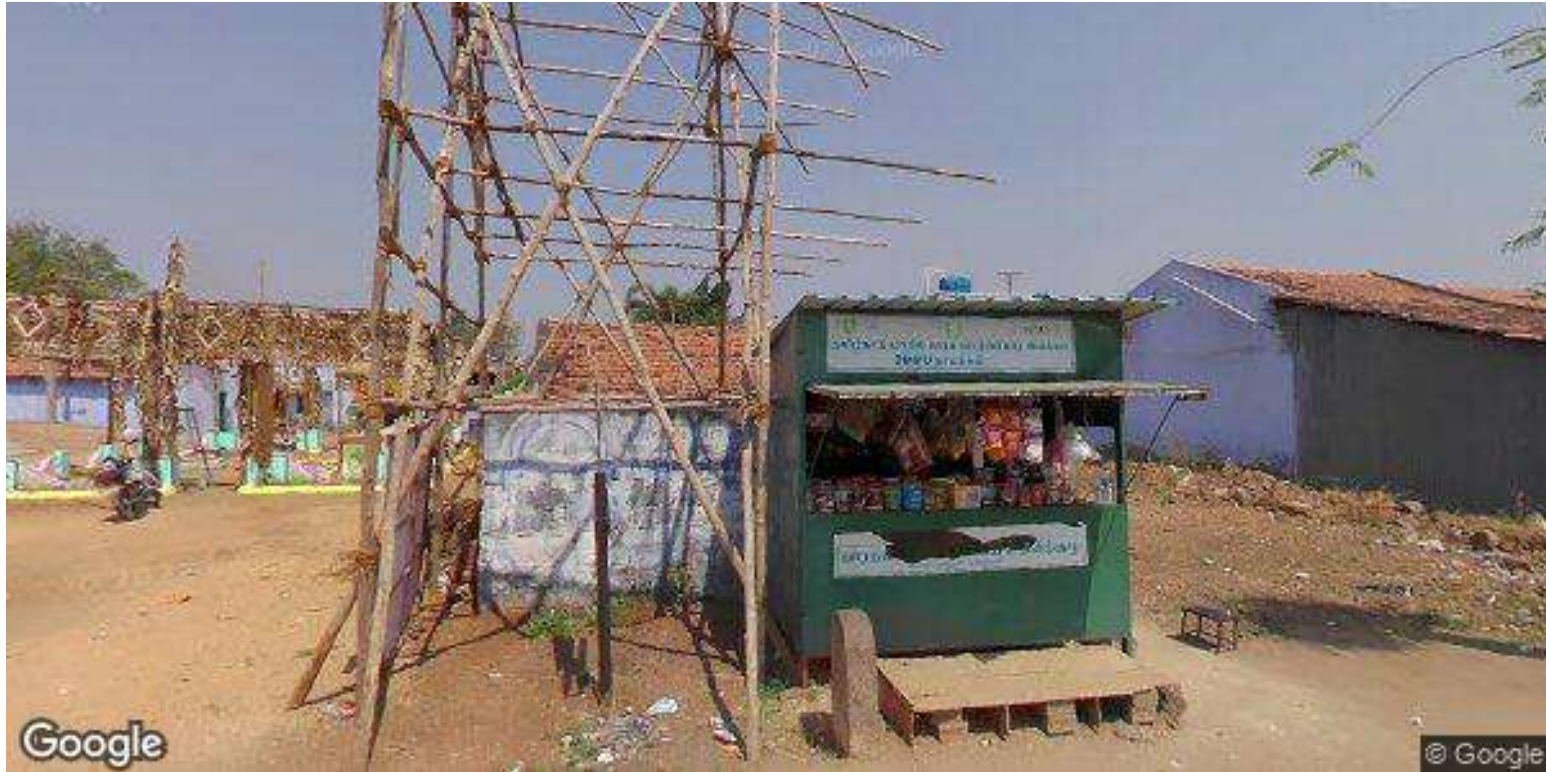


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47426, 78.08763



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.47540, 78.08765



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47566, 78.08963

